

A multi-loop auto-locking system with fully digital electronics for ultra-stable laser

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To achieve high performance ultra-stable laser (USL), multiple locking loops are required to operate simultaneously, including such as Pound–Drever–Hall (PDH), fiber noise cancellation (FNC), light power stabilization (LPS). In this paper, a highly-integrated multi-loop auto-locking system designed for USLs is demonstrated using fully digital electronics. The auto-locking process is implemented using a field programmable gate array with high-speed and high precision analog-to-digital and digital-to-analog converters. The auto-locking system can automatically stabilize the laser carrier frequency or offset sideband frequency to its reference cavity, and is capable of successfully relocking within 10 s if interrupted. The fully-digital open-loop bandwidth of the fast feedback loop reached 1.3 MHz. The frequency instability of the USL's frequency in carrier- and offset-locking modes was measured to be 1.5×10^{-15} at 1 s and 2.5×10^{-15} at 1 s, respectively. The contribution of digital electronics to the instability in carrier-locking mode was 2.9×10^{-16} at 1 s, while the noise suppression effect of the digital servo inside the FNC and LPS loops reached 75 dB and 40 dB at 1 Hz, respectively. The results show that this highly flexible, and high-reliability digital auto-locking system can fully satisfy the requirements of USL applications.

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I. INTRODUCTION

As the next generation of laser frequency standards, ultra-stable lasers (USLs) have the advantages of narrow linewidth and good short-term stability. As a result, USLs play a crucial role in cutting-edge research and precision measurement fields, such as optical clocks¹, gravitational wave detection², earth gravity field measurement³, and time-frequency transfer⁴. In these fields, the noise level of the USLs directly affects measurement accuracy, and the best USL stability levels demonstrated under laboratory conditions currently reach the 10^{-17} level⁵⁻⁷. However, as the number of engineering projects requiring precision measurements increases, the need for transportable USLs has emerged, with many recent research studies conducted in this field⁸⁻¹². For transportable USLs, especially those intended for space missions, the system volume needs to be minimized as much as possible, and integration and automation need to be improved, under the constraint of frequency and power stability.

Over the past decade, there have been many studies on the development of auto-locking servos for USLs. The hybrid analog digital circuit scheme^{13,14} realizes servo calculations via analog circuits, while using digital circuits for parameter changes and automation control. This scheme achieves low noise due to the high bandwidth of the analog circuits utilized, but its flexibility and integration options are limited by the number of gain adjustment switches or digital potentiometers. With the development of digital servos in various laser fields¹⁵⁻¹⁷, various digital auto-locking systems have been applied for the implementation of single Pound-Drever-Hall (PDH) locking loops^{18,19}. Furthermore, due to the requirement of maintaining the USL's performance under various conditions, multiple locking loops, including PDH, fiber noise cancellation (FNC), and light power stabilization (LPS) etc., should be applied simultaneously to maintain the frequency stability of the USL. The use of multi-loop auto-locking systems also helps increase the system's robustness and adaptability to harsh environments. Therefore, the study of auto-locking systems is a very important and challenging for multi-loop USLs.

In this paper, a locking system with all loops controlled using fully digital servos is proposed as a design optimized for high integration, high bandwidth operation and extensive control function flexibility. The implementation is realized on a single FPGA processor. In this study, a USL system using an acousto-optic modulator (AOM) for fast PDH feedback

is considered. Since the gain ripple of the AOM's radio-frequency (RF) driving circuit is caused by different frequency signals or temperature influence, the LPS loop is coupled with the PDH fast feedback loop to suppress the power fluctuations of both the laser and the AOM's circuit. An improved version of the auto-locking algorithm is also proposed to ensure successful USL auto-locking regardless of inappropriate initial parameters caused by drifts in the light power. The proposed auto-locking algorithm not only supports locking of the USL's carrier frequency, but also of the offset sideband frequency, which is important for many applications such as frequency shifting locking of different lasers^{20,21}. The proposed USL system can correctly identify the carrier signal and two offset sideband signals, and automatically stabilize one of them to its reference cavity. For this application, the control sequence of the locking loops is LPS, PDH, LPS, FNC in order. The LPS loop is controlled twice - first for coarse adjustment to maintain constant laser power input before each PDH loop locking, thus ensuring stable initial conditions for the PDH loop which has the most complex active locking procedure, and then to stabilize the laser power fluctuation after PDH locking. Finally, the FNC loop is controlled to reduce additional noise imposed by the optical fiber on the output laser.

II. DESIGN AND ANALYSIS

A. Digital Servo design

As shown in Fig. 1, the locking system mainly consists of three loops for stabilizing the USL's frequency (PDH), phase (FNC) and power (LPS). In the PDH loop, the output of a 1550 nm fiber laser is frequency-modulated using an AOM1, and then a portion is split into a broadband EOM by the 1×2 optic coupler OC1. The electronic sideband technique^{22,23} is used to cascade the phase modulation of light using the EOM. The modulation structure is shown in the red dashed box in Fig. 1, where ω_c and ω_{gap} are the carrier and offset sideband frequencies, respectively. Locking the first-order sideband with a frequency of $\omega_{gap} \pm \omega_{PDH}$ on the F-P cavity allows the adjustment of the USL's frequency over a wide range, and ω'_{PDH} generated by the frequency synthesizer RF1 is the demodulation frequency used to obtain the PDH error signal. The modulation frequency ω_{PDH} and the shifting frequency ω_{gap} are both generated by the frequency synthesizer RF2 and transferred to electro-optical

modulator (EOM). The modulation frequency ω_{PDH} and the demodulation signal ω'_{PDH} have the same frequency with a fixed phase difference, with the 10 MHz Rb clock acting as a reference source. Then, the output of the EOM is processed using a free-space optical path and reflected by the F-P cavity. The reflected light signal of the F-P cavity is detected by photo-detector PD1 and divided into AC and DC parts using a bias-tee. The AC part is demodulated by the mixer, filtered by the low pass filter (LPF), and sent to the digital servo processing unit, as shown on the bottom right of the figure. The processing unit comprises the PDH analog-to-digital converter (ADC), a field-programmable gate array (FPGA), and the PDH digital-to-analog converters DAC1 and DAC2. The digital servo's output is used to adjust the driving voltage of the voltage-controlled oscillator VCO1 and a piezo-transducer (PZT). The DC signal is collected by the LPS' ADC for determining the PDH loop's locking status. The modulation and demodulation signals in the PDH loop are obtained from the signal source to achieve signal synchronization and guarantee that the PDH error signal is at its maximum value.

The other output of OC1 is sent through OC2 to the FNC loop. One part reflected by the Faraday mirror (FM) serves as the reference light, while the other part carries twice the fiber noise information through the fiber and AOM2. After beating with the reference light, the noise information is sent to PD2, the output of which passes through a phase and frequency detector (PFD) to generate an error signal. The error signal is then passed through a separate processing pathway of the digital servo, composed of the FNC ADC, the FPGA pathway, and the FNC DAC, and is finally fed back to VCO2. Both VCO1 and VCO2 are equipped with adjustable offset voltages (controlled using two additional DACs) to maintain the output frequency of the VCOs near the center frequencies of the different AOMs. The digital servo and AOM2 in the FNC loop can also be used to lock the USL to the optical comb, through the switching of PD3's beat-note signal between the USL light and the optical frequency comb to the PFD and modulating the reference clock ω_{FNC} to generate a suitable error signal for the locking loop.

In the LPS loop, the DC signal of PD1 is processed using a digital servo pathway comprising an LPS ADC, an FPGA pathway, and an LPS DAC, and then fed back to the voltage variable attenuator (VVA) to adjust the AOM's RF driving power in the PDH loop and stabilize the AOM's output optical power. The AD/DA used for the three loops are shown in Table 1.

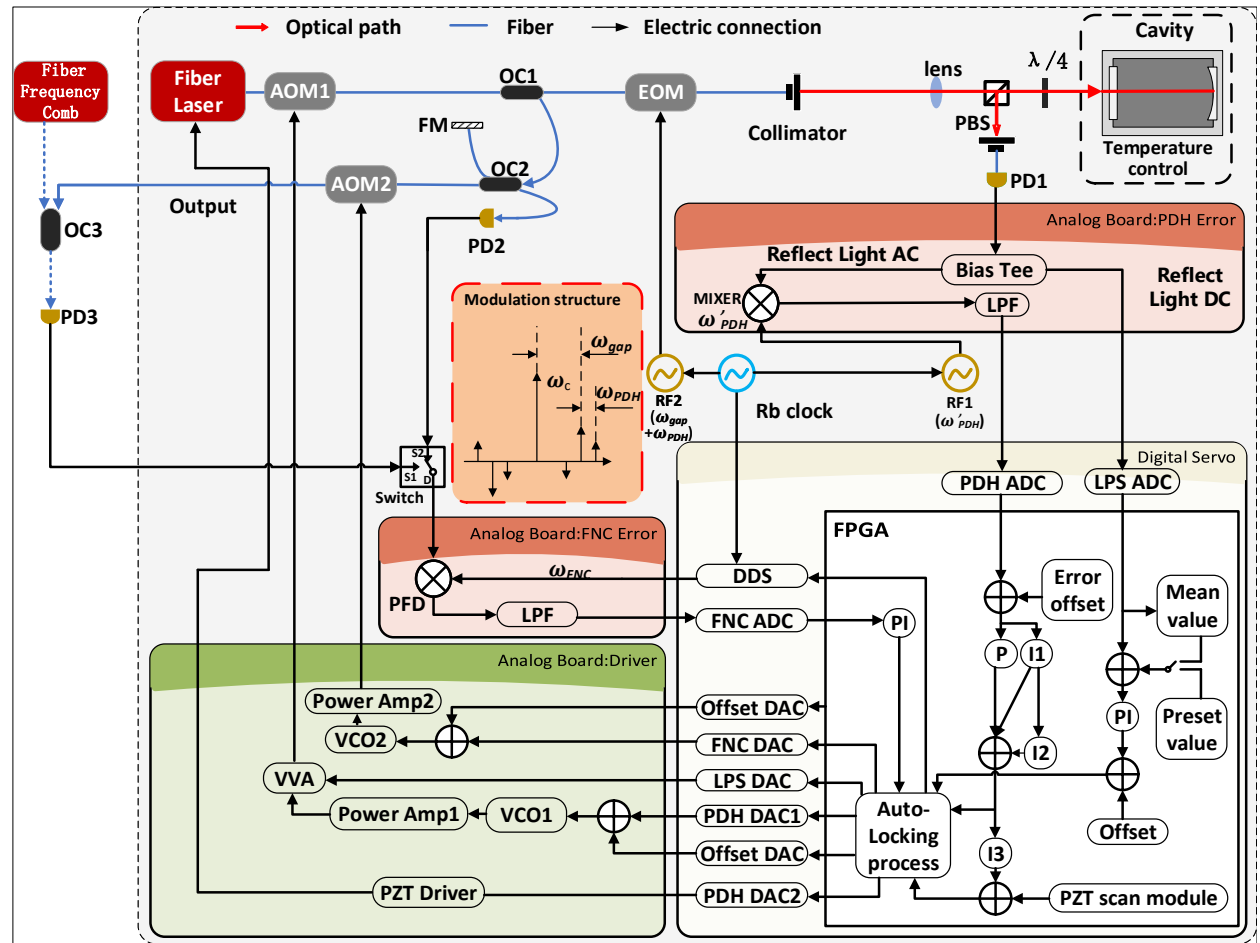


FIG. 1. Schematic of the USL's digital locking system. PZT: piezo-transducer; OC: polarization-maintaining fiber optic coupler; FM: Faraday mirror; $\lambda/4$, quarter-wave plate; EOM: electro-optical fiber ; PBS: polarization beam splitter; PD: photo-detector; VCO: voltage-controlled oscillator; LPF: low-pass filter; DDS: direct digital synthesis; LPS: light power stabilization; FNC: fiber noise cancellation; VVA: voltage variable attenuator; PFD: phase and frequency detector; PI: proportional-integral controller; ω_c : carrier frequency; ω_{gap} : tunable modulation frequency; ω_{PDH} : fixed modulation frequency; and ω'_{PDH} : fixed demodulation frequency.

TABLE I. AD/DA Paramters Applied in the Three Loops.

Chip Resolution	Sampling Rate
(PDH/FNC/LPS)	(PDH/FNC/LPS)
ADC 16 bit	100 Msps/93 ksps/93 ksps
DAC 16 bit	100 Msps/465 ksps/465 ksps

For fast PDH feedback, a PID controller is adopted in the form of $P + I1 + I2$, while for slow feedback, it is $(P + I1 + I2)I3$, with multiple integration circuits to suppress the Hz-scale frequency flickering and random walk frequency noise of the laser²⁴. For the other locking loops, PI controllers are used to suppress phase or voltage noise.

For the fast PDH loop, the open-loop transfer function is given by:

$$H_o = \frac{\pi \cdot f_{cav}}{s + \pi \cdot f_{cav}} \cdot D \cdot \left(K_p + \frac{K_{I1}}{s} + \frac{K_{I2}K_{I1}}{s^2} \right) \cdot K_{vco1} \quad (1)$$

where f_{cav} is the linewidth of the cavity, and $D, K_p, K_{I1}, K_{I2}, K_{vco1}$ represent the frequency discrimination coefficient, the proportional gain, the first- and second-order integration gains, and the VCO1 gain, respectively.

For the FNC loop, the open-loop transfer function is:

$$H_o = D_2 \cdot \left(K_{fnc-p} + \frac{K_{fnc-I}}{s} \right) \cdot \frac{K_{vco2}}{s} \quad (2)$$

where D_2 is the phase detection coefficient, and $K_{fnc-p}, K_{fnc-I}, K_{vco2}$ represent the frequency discrimination coefficient, the proportional gain, the first-order integration gain, and the VCO2 gain, respectively.

For the LPS loop, the transfer function is:

$$H_o = \frac{2\pi \cdot f_{Bias-tee}}{s + 2\pi \cdot f_{Bias-tee}} \cdot G_0 \cdot \left(K_{lps-p} + \frac{K_{lps-I}}{s} \right) \quad (3)$$

where $f_{Bias-tee}$ is the cut-off frequency of the bias-tee DC output, and K_{lps-p}, K_{lps-I} represent the proportional gain and the first-order integration gain, and G_0 is the gain without the PID feedback.

All PID operations of the locked loop are performed inside the FPGA, which has a clock of 100 MHz. All P and I modules use first-order infinite impulse response (IIR) filters with adjustable gain and different switch controls. Similar filters can be found in [17]. In the design, the P parameter is adjusted within the range of -50 dB to +50 dB with resolution of -80 dB, while the I parameter can set from around 1 Hz to 400 kHz with 0.2 Hz resolution, which meets practical requirements.

B. Auto-locking logic design

The logic diagram for the auto-locking system is illustrated in Fig. 2 and is divided into five stages. In Stage I, upon receiving the start command from the main computer, the FPGA first checks the telemetry status. It then determines the USL mode, i.e. whether it is operating as a reference source in the carrier-locking mode or as a frequency bridge in offset-locking mode, and initializes the PID parameters for both modes. The system then activates the LPS PID to correct the initial offset of the LPS DAC, and the DAC output voltage changes the RF input power of AOM1 by adjusting attenuation of the VVA to realize the light power stable around the preset value. This approach ensures that the total gain of the loop, excluding the PDH servo, remains unchanged, allowing the system to automatically lock using an appropriate default trigger threshold and optimal default PI parameters. This prevents situations where the preset trigger threshold and PI parameters are unsuitable, which would make locking impossible.

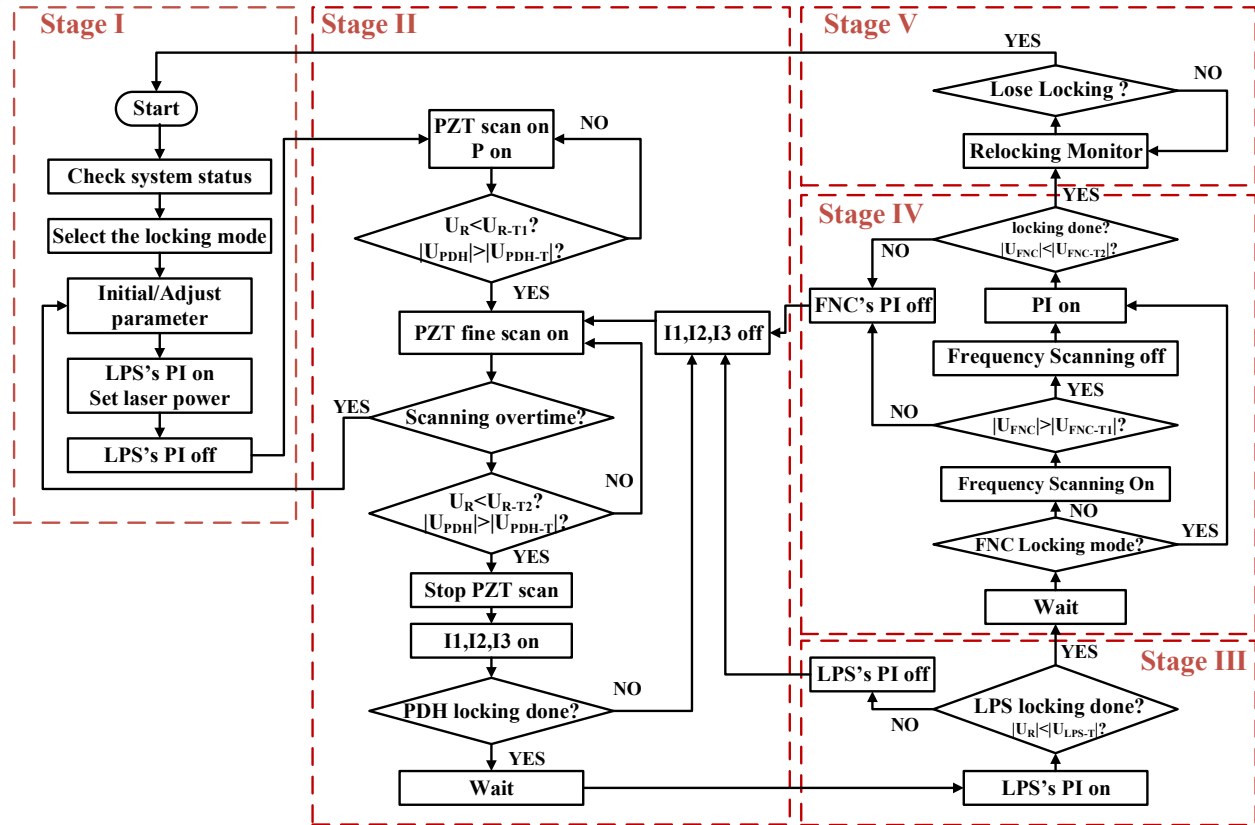


FIG. 2. Logic block diagram of automatic laser frequency locking.

In Stage II, the FPGA controls the PZT voltage to scan the laser frequency over a

wide range and uses the LPS ADC to obtain the reflected light's DC signal U_R to search for resonance peaks and determine the locking state. During the laser frequency scanning process, the PDH PID remains in the P state, facilitating the search for resonant peaks. To unify the judgment conditions for the carrier and the offset-locking modes, an error signal judgment condition is introduced. When the peak of the DC signal is smaller than a threshold U_{R-T1} and the error peak $|U_{PDH}|$ is larger than $|U_{PDH-T}|$, PZT conducts fine scanning to find the optimal lock-in moment by gradually narrowing the scanning voltage range until the DC signal reaches the threshold U_{R-T2} . The PDH PID components I1, I2, and I3 will be immediately enabled after PZT fine scanning is concluded. PDH locking will be performed as the DC and error signals are maintained within the triggering threshold range; otherwise, the PID of the PDH will be disabled to restart PZT voltage scanning, and the system will return to Stage I to adjust the laser parameters for relocking if the allowed scanning time has been exceeded.

In Stage III, the FPGA transitions to the LPS stage, at which time it will first calculate the average value of U_R , and then activate the LPS PID to stabilize U_R around the average value U_{LPS-T} . If the system briefly loses lock of the laser due to some adverse event, such as an impact, without the LPS servo, U_R would exceed U_{R-T2} . However, due to the coupling of the PDH locking loop and the LPS loop, U_R will still be maintained near its previous average value. The criterion for determining whether LPS lock has been lost is whether the output of the LPS PID exceeds the set threshold. When LPS unlock occurs, the PIDs of the LPS and PDH are turned off, and then a fine scan of the laser frequency is performed. If the scan times out and relocking has not been achieved, this indicates that the system is being interfered with by other factors or the light power drift is causing inappropriate locking parameters, preventing the locking process. In this case, the system will eventually return to Stage I, adjust the threshold value and PID parameters and restart the locking process.

In Stage IV, if the USL is used as a reference source, the locking process of the FNC loop is relatively simple. First, the PID is turned on and then the average value of the error signal U_{FNC} is calculated. If the result meets the U_{FNC-T2} threshold requirements, the FNC loop is successfully locked; otherwise, the system will turn off the FNC PID and return to Stage II to find the appropriate locking parameter. When the FNC loop acts as the frequency shifting locking loop, the system will check the optical comb status during the configuration step in Stage I. An additional frequency scanning is then applied on the EOM to determine

the proper error signal which is larger than the threshold U_{FNC-T1} , and the same process is carried out in the FNC loop to complete the locking process.

After all the loops have accomplished locking, the relocking monitor in Stage V is put in operation to monitor the system locking status. If the system is unlocked, it will automatically return to Stage I to restart the locking flow.

III. RESULTS AND DISCUSSION

A. Bandwidth

We measured the -180° phase shift bandwidth of the three loop digital servos (PDH, LPS, FNC) with only the proportional (P) control enabled, as shown in Fig. 3(a). The measured bandwidths were 1.3 MHz, 30.2 kHz, and 30.2 kHz, respectively. Based on the relationship between the -180° phase shift bandwidth f and the delay $\tau = 1/2f$, the delays for the three loops were calculated to be 384 ns, 16.6 μ s, and 16.6 μ s.

All three transfer functions exhibited a 90° phase shift in the high-frequency range, indicating that the remaining 90° phase shift may be attributed to loop delay. The delays for the other components of these three loops, which included frequency discrimination (210 ns for PDH loop and 2.47 μ s for FNC loop), the VCO (120 ns for both PDH and FNC loop) and the AOM (0.5 μ s for both PDH and LPS loop, 1 μ s for FNC loop), resulted in total delays of 1.2 μ s (PDH loop), 20.2 μ s (FNC loop), and 17.1 μ s (LPS loop), respectively. The closed-loop bandwidth was estimated using the equation $f = 1/4\tau$, yielding values of 206.6 kHz, 12.4 kHz, and 14.6 kHz for the respective loops.

In order to measure the closed-loop bandwidth of the PDH loop, the reflected light AC signal was split using a power divider and measured its spectrum by a spectrum analyzer. Then, the P gain of the PID was gradually increased, which led to servo pumping. The difference between the servo pump and the center frequency can be approximated as the closed-loop bandwidth. Of course, this is not a stably locked state. From Fig. 3(b), it can be seen that the closed-loop bandwidth of the system was about 200 kHz. Similarly, bandwidths of approximately 12 kHz (see Fig. 3(c)) for the FNC loop and 13.6 kHz (Fig. 3(d)) for the LPS loop were obtained, which are close to the estimated closed-loop bandwidths.

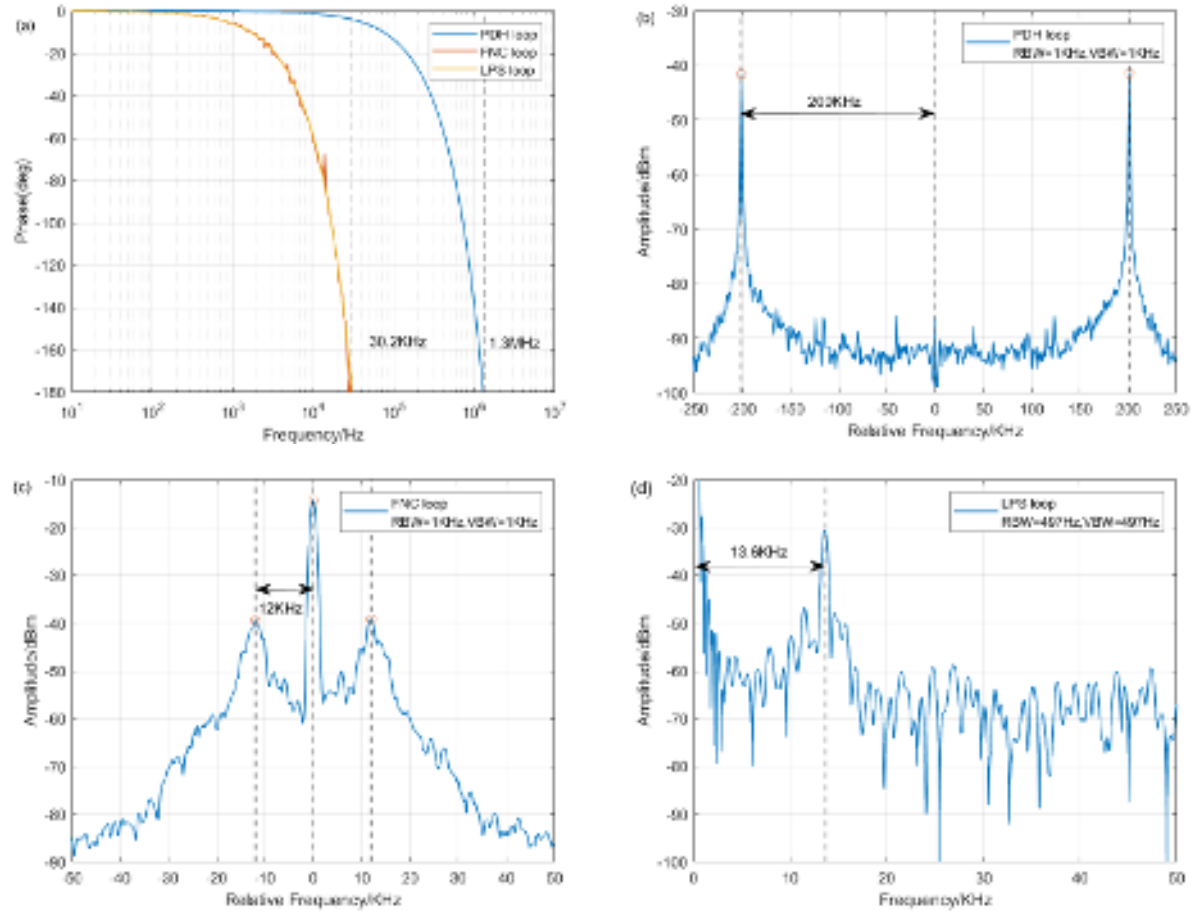


FIG. 3. (a) Open loop bandwidth of the digital servos in three loops. (b) The closed-loop spectrum of AC signal of PD1 in PDH loop. (c) The closed-loop spectrum of PD2 output in FNC loop. (d) The closed-loop spectrum of U_R .

B. Auto-locking

The measured results of auto-locking and relocking for the carrier-locking mode, are shown in Fig. 4. At -2.38 s, the FPGA received a lock start command from the main computer and immediately activated the PID of the LPS, adjusting the amplitude of the reflected light's DC signal to 450 mV. Then, a wide-range laser frequency scan was initiated. At -0.25 s, $U_R < U_{R-T1}=400$ mV, so the scan stopped, and the PID of the PDH lock loop was turned on, quickly completing the lock. After that, the LPS PID also went into operation, and the FPGA entered the lock monitoring state. At 0.43 s, the RF drive of the EOM was manually turned off as an interference, and the PI output of the LPS quickly

exceeded the set threshold. The FPGA judged that the system lost lock and turned off the PIDs of the LPS and the PDH. U_R also returned to its pre-lock level, and then the FPGA entered the relocking state to perform a fine scan of the laser frequency. During this period, although the carrier frequency passed through the resonant frequency of the F-P cavity multiple times, the system determined that the disturbance had not ended since $|U_{PDH}| < |U_{PDH-T}|$, so the scan would not stop. The EOM's RF drive was turned on again at 1.24 s, and at 1.37 s, the laser frequency approached the resonance frequency, completing the locking process again.

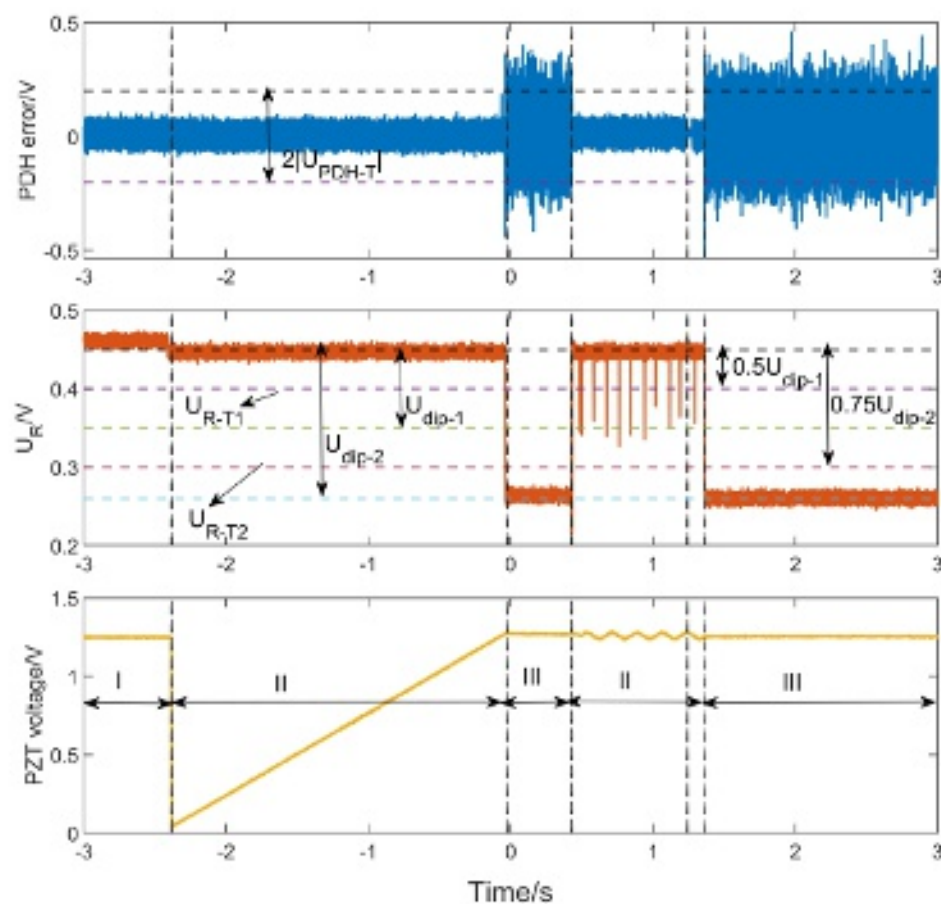


FIG. 4. PDH error, reflection light DC (U_R) and PZT voltage during auto-locking and relocking for carrier-locking mode.

As shown in Fig. 5, in the carrier-locking mode, the configuration of the P state deepens and widens the U_R dip of the carrier at the resonance frequency, while making the U_R dip of the sideband at the same frequency shallower and narrower. In offset-locking mode, the

U_R dip of the target offset sideband at the resonance frequency also becomes deeper and wider, while the other offset sideband's U_R dip at the same frequency becomes shallower and narrower. This ensures that the FPGA can accurately identify the correct offset sidebands, as the power of the other sidebands is sufficiently low to avoid interfering with the search process.

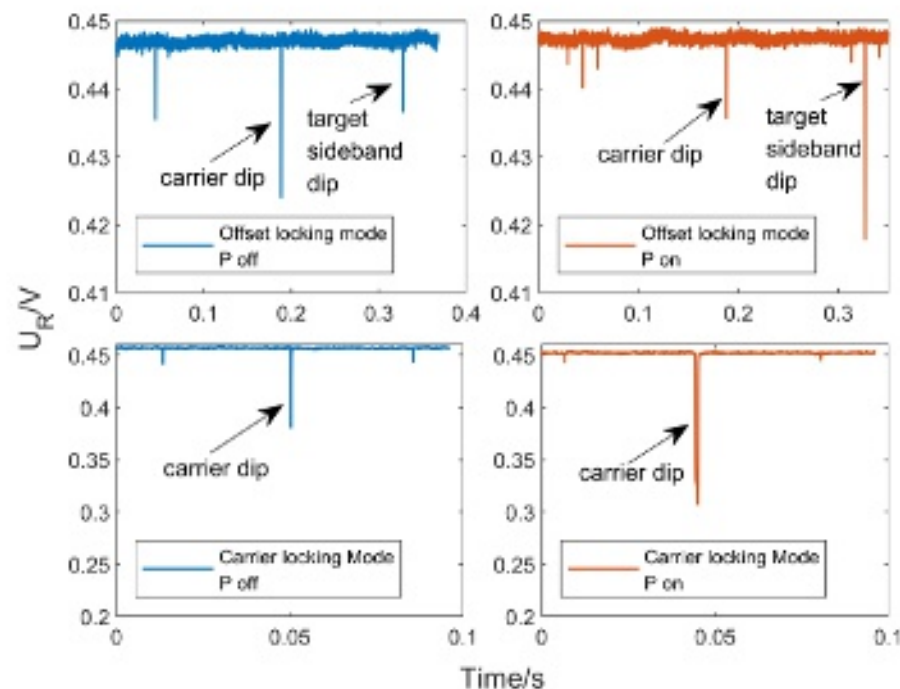


FIG. 5. U_R 's dip in different operation modes.

For the carrier-locking mode, when U_R falls below the triggering threshold U_{R-T1} , which is set to the maximum value of U_R minus 50% of U_{dip-1} , this indicates that the carrier's optical frequency is approaching the resonance point. In offset-locking mode, due to the similarity between the dips of the carrier's U_R and the target offset sideband's U_R , the absolute amplitude of the PDH error signal $|U_{PDH}|$ is used to distinguish the DC signals in different modes. When the error signal exceeds the triggering threshold $|U_{PDH-T}|$ (which is twice the noise level of the PDH error signal), it can be concluded that the target offset sideband frequency is near the resonant peak; at that point, the carrier will not generate an error signal, so the system will not be confused.

Compared to carrier-locking mode, the U_R dip in offset-locking mode is much smaller; when the average U_R value is 450 mV, the dip is only about 30 mV. A U_R drift of several tens

of mV can make the preset trigger threshold no longer suitable, making it difficult to lock. Therefore, when the lock time exceeds expectations, the light power needs to be recalibrated. When the laser is used for frequency shift locking between two lasers of different wavelengths, the frequency scanning on the EOM will cause a relock of the system. The algorithm has great reliability, and the digital system relocked successfully within 10 seconds over 30,000 times by monitoring the changes of the PDH error and DC signals.

C. Instability of beatnote in PDH loop

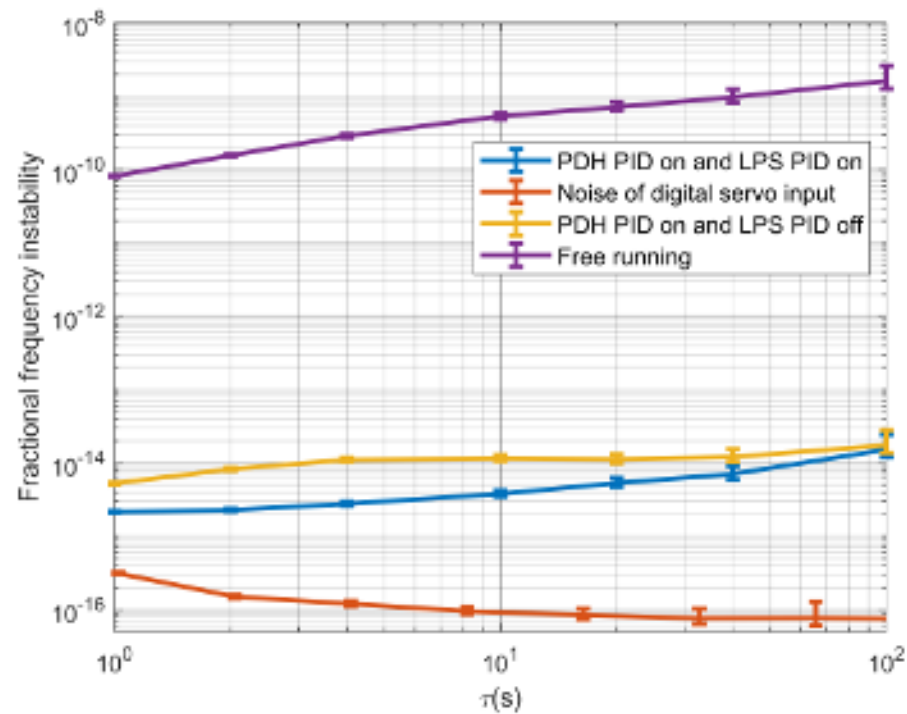


FIG. 6. The blue, red, and purple lines represent the allan deviation of the beat note between Laser 1 in the carrier locking mode with Laser 2. The red line represent the noise of digital servo input which has been transferred into fractional frequency instability from voltage.

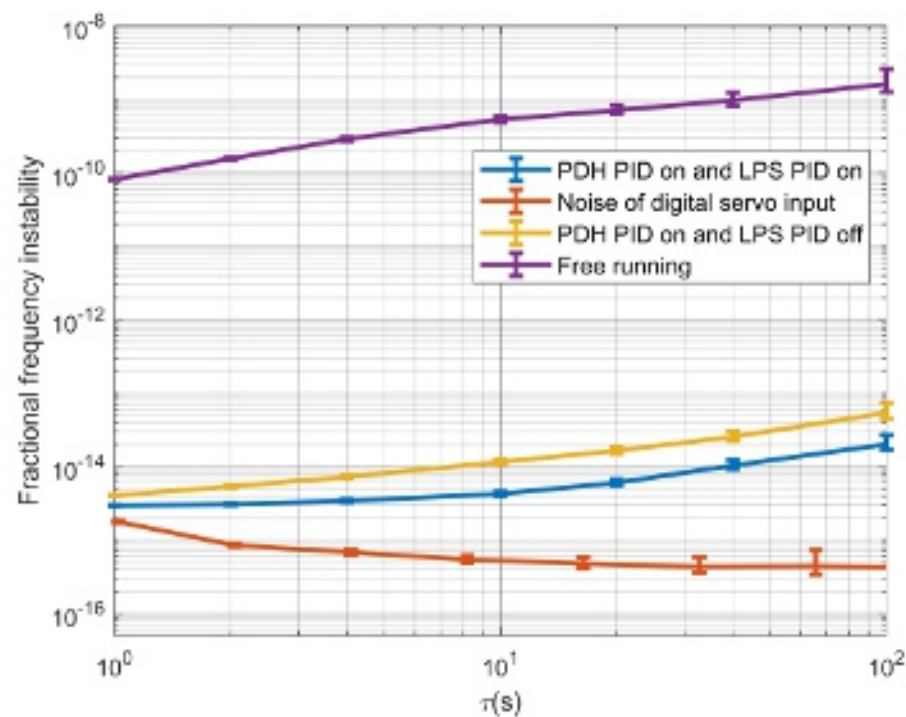


FIG. 7. The allan deviation in offset locking mode. The beat note is measured by a frequency counter. The noise of digital servo input is measured by PDH ADC.

For the USL's performance indicator assessment, our main focus is on its low-frequency noise level. For PDH loops, the input noise of the digital servos cannot be suppressed using high gain loops and is the main source of noise in the circuit. In Fig. 6 and Fig. 7, a frequency discrimination coefficient was used to convert servo input noise into frequency noise for comparison with the free-running noise and locked noise of the laser. To test the performance of the fully digital auto-locking laser system, Laser 1 was beat before and after locking with another 1550 nm USL (Laser 2), which was locked with a similar performance of around 2.5×10^{-15} at 1 s. The beat stability before locking was 8.1×10^{-11} at 1 s, mainly contributed by Laser 1; after the PDH loop of Laser 1 was locked, the beat frequency instability with the LPS PID turned off in carrier-locking mode was 5.3×10^{-15} at 1 s. After the activation of the LPS PID, the beat frequency instability decreased to 2.2×10^{-15} at 1 s. The stability of Laser 1 was calculated to be approximately $= \frac{2.2 \times 10^{-15}}{\sqrt{2}} = 1.5 \times 10^{-15}$ at 1 s. The contribution of the digital servo's input noise to instability was measured approximately 2.9×10^{-16} at 1 s, and therefore, the instability of the beat frequency was mainly limited by the noise of the cavity and other noises in the optical path, rather than circuit noise.

In offset-locking mode, the low sideband power resulted in a reduced frequency discrimination coefficient, leading to an increased contribution of the digital servo's input noise to instability. The beat frequency instability in offset-locking mode was measured results 2.9×10^{-15} at 1 s in the blue curve of Fig. 7. Considering the instability of Laser 2 is around 1.5×10^{-15} at 1 s, the instability of Laser 1 can be calculated to be approximately $\sqrt{(2.9 \times 10^{-15})^2 - (1.5 \times 10^{-15})^2} = 2.5 \times 10^{-15}$ at 1 s. The contribution of the digital servo's input noise to instability reaches 1.7×10^{-15} at 1 s. The noise contribution of electronics can be reduced through an improvement of the frequency discrimination coefficient or through an increase of the light's power. However, due to the limitation of the ADC's input range, the light power cannot be increased beyond a certain point, which makes it difficult to define the threshold required for the automatic locking process.

D. Phase noise of FNC Loop

The performance of the digital FNC circuit was measured. Figure 8 shows the phase noise under open-loop (blue) and closed-loop (yellow) conditions, as well as the theoretically predicted (red) and the RF reference noise output by the DDS (purple). The theoretical prediction noise was obtained by multiplying the phase noise with the FNC PID turned off and the noise transfer function Eq. (2). As shown in Fig. 8, the phase noise with the FNC PID turned on between 35 Hz and 1 kHz closely matches the theoretically predicted noise levels, with the phase noise suppression effect determined by the loop gain. However, due to the loop's bandwidth constraint, further increasing the loop gain could destabilize the system. To suppress noise further, faster ADCs and DACs need to be employed to reduce loop delay and enhance both loop gain and bandwidth. For the FNC loop, the primary focus was on the level of low-frequency noise suppression, and the current bandwidth fulfilled this requirement adequately. If the FNC loop was used to lock the USL onto the optical comb, the bandwidth even needs to be reduced to avoid interference from the optical comb on the short-term stability of the USL. From 0.1 Hz to 35 Hz, the phase noise with the FNC PID turned on was limited by the RF reference phase noise from the DDS output, preventing it from being reduced to the theoretical level. The phase noise suppression at 1 Hz reached approximately 75 dB. According to the experiment results, the contribution of fiber noise to instability was 1.3×10^{-17} at 1 s, which can be ignored.

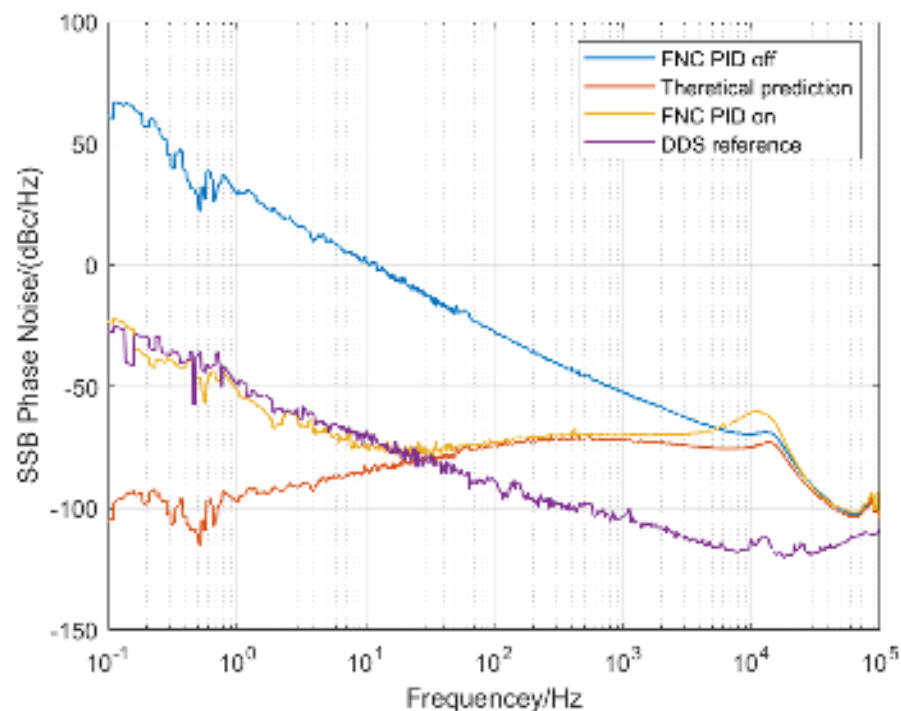


FIG. 8. The single sideband phase noise measured with the phase noise analyzer.

E. In-loop noise of LPS loop

The performance of the LPS loop, was measured next. Figure 9 displays the light power noise under open-loop (blue) and closed-loop (yellow) conditions, as well as the theoretically predicted (red) and ADC (purple) noise. The theoretically predicted noise was obtained through the multiplication of the open-loop noise with the noise transfer function Eq. (3). The data were collected from the LPS ADC inside the loop. As the digital servo did not have a high-speed data interface, the raw data were exported at a lower sampling rate after low-pass filtering inside the FPGA. Therefore, the upper limit of the horizontal axis is 1 kHz. In LPS loops using analog servos, the noise inside the closed-loop is theoretically limited only by the reference voltage noise²⁵. In the proposed LPS digital servo loop, the noise of the ADC is much higher than the reference voltage noise, and is the determinant factor in the closed-loop loop noise, and cannot be reduced to the same level as the theoretically predicted noise. Therefore, the noise at 1 Hz can be suppressed by about 40 dB.

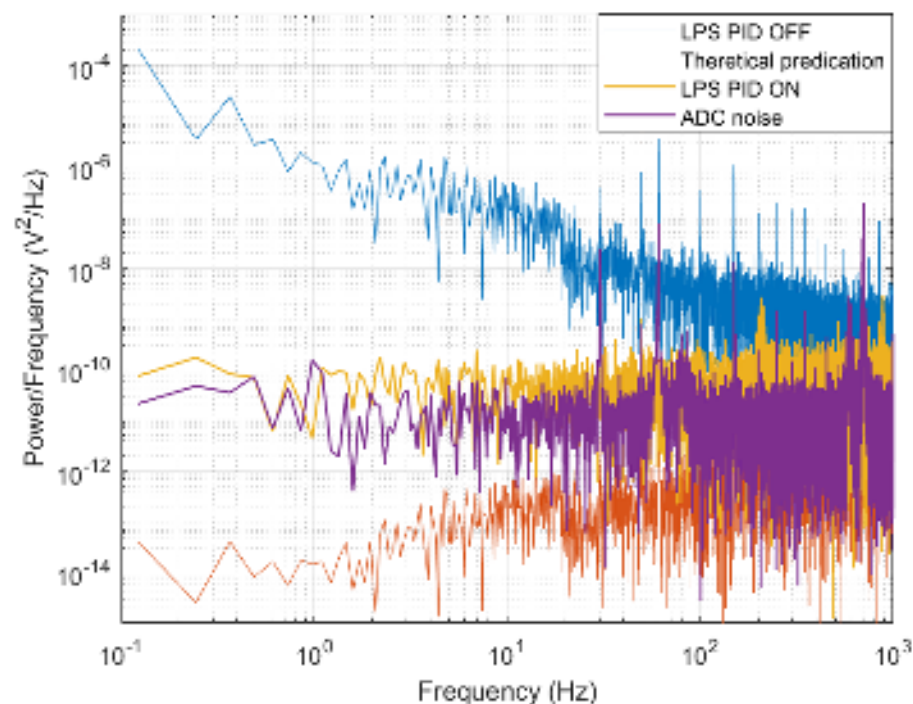


FIG. 9. In-loop noise of LPS measured by LPS ADC. ADC noise(purple curve) is measured by short circuiting the ADC input.

IV. CONCLUSION

In this paper, a highly-integrated fully digital auto-locking electronics setup designed for USL systems is discussed. The digital system bandwidths of the PDH, LPS and FNC loop are 1.3 MHz, 30.2 kHz, and 30.2 kHz, respectively. In addition, the auto-locking algorithm is designed in a manner which ensures that the auto-locking of USLs will not fail due to inappropriate default parameters caused by changes such as light power. The system successfully auto-locked and relocked the laser frequency over 30,000 times with great reliability. The frequency instabilities of the USL frequency in carrier- and offset-locking modes were measured to be 1.5×10^{-15} at 1 s, and 2.5×10^{-15} at 1 s, respectively. At 1 Hz, the suppression effect of FNC and LPS loops on the noise inside the closed loop reached 75 dB and 40 dB, respectively.

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